Lab 14 – (OEL)
Database Application Using Python

Programming Exercise:

Task#1.

Design a database having at least two tables using any DBMS of your own choice.
1. Dept (deptid(pk),dname)

2.Teacher (Tid(pk),name,designation,salary,deptid(fk))

3. Courses (ccode,cname,credithrs,Tid(fk))

In this lab, student will learn how to make a Database App using Python

● Create database connection

● Design GUI ofyour application according to your own choice using Tkinter or Custom Tkinter Module.

● Write Python code to display data from database into python program. DML(Insert& Delete)

Task#2.

● Write a Python code to perform insert and delete operation using SQL queries on at least one table mentioned above. When user clicks on delete button , it deletes a particular record and display a message that a record has been deleted successfully.

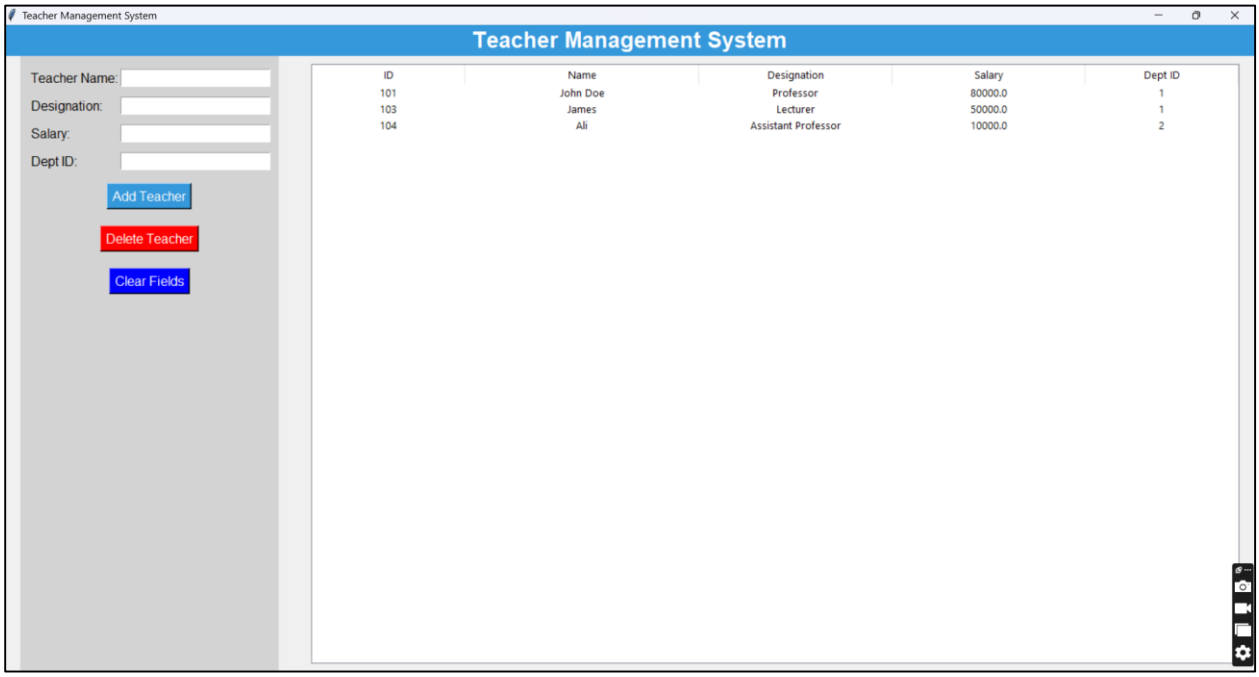
Source Code:

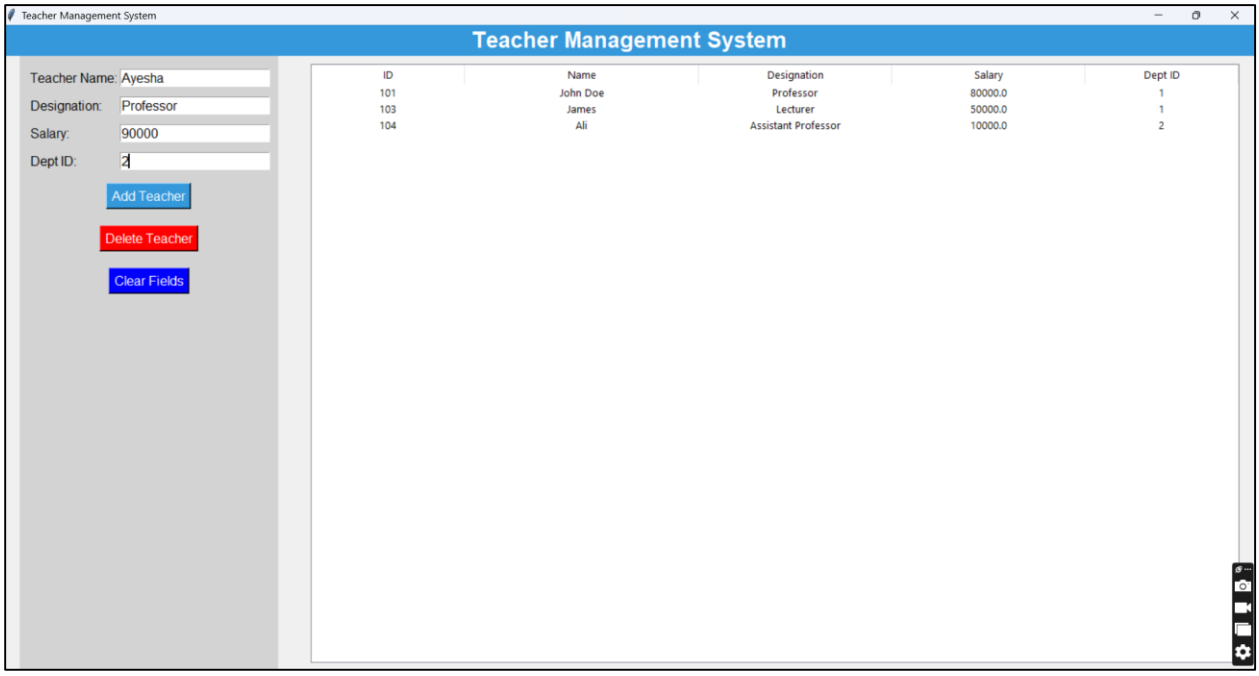
```python
import sqlite3
from tkinter import *
from tkinter import ttk
from tkinter import messagebox

# SQLite database connection
def connect_db():
    conn = sqlite3.connect('university.db')
    return conn


# Create tables in the database
def create_tables():
    conn = connect_db()
    cursor = conn.cursor()

    # Create Dept table
    cursor.execute('''CREATE TABLE IF NOT EXISTS Dept (
        deptid INTEGER PRIMARY KEY AUTOINCREMENT,
        dname TEXT NOT NULL)''')

    # Create Teacher table
    cursor.execute('''CREATE TABLE IF NOT EXISTS Teacher (
        Tid INTEGER PRIMARY KEY AUTOINCREMENT,
        name TEXT NOT NULL,
        designation TEXT NOT NULL,
        salary REAL NOT NULL,
        deptid INTEGER,
        FOREIGN KEY(deptid) REFERENCES Dept(deptid))''')

    conn.commit()
    conn.close()


# Display data from the Teacher table
def display_data():
    for item in teacher_table.get_children():
        teacher_table.delete(item)

    conn = connect_db()
    cursor = conn.cursor()
    cursor.execute('''SELECT Tid, name, designation, salary, deptid FROM Teacher''')
    rows = cursor.fetchall()

    for row in rows:
        teacher_table.insert("", "end", values=row)

    conn.close()


# Add a new teacher
def add_teacher():
    try:
        conn = connect_db()
        cursor = conn.cursor()
        cursor.execute("INSERT INTO Teacher (name, designation, salary, deptid) VALUES (?, ?, ?, ?)",
                       (var_teacher_name.get(), var_teacher_designation.get(),
float(var_teacher_salary.get()), int(var_teacher_deptid.get())))
        conn.commit()
        conn.close()
        messagebox.showinfo("Success", "Teacher added successfully")
        clear_inputs()
        display_data()
    except Exception as e:
        messagebox.showerror("Error", f"Failed to add teacher: {e}")
```

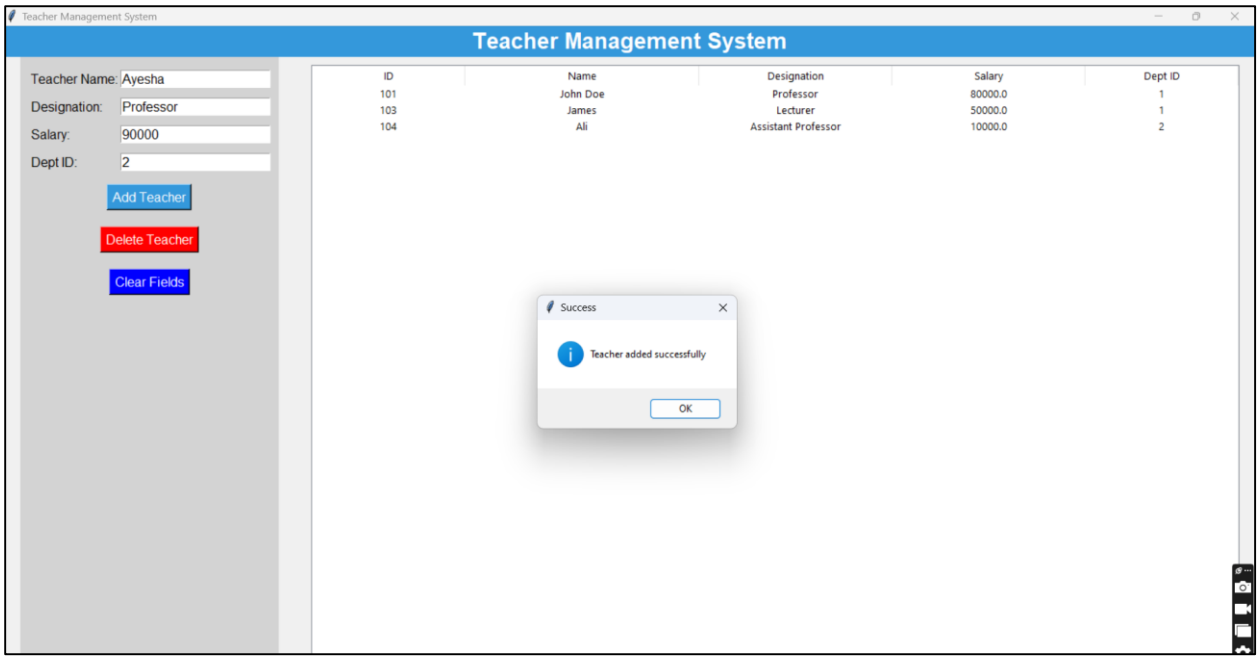
```python
# Delete a teacher
def delete_teacher():
    try:
        selected_item = teacher_table.selection()
        if not selected_item:
            messagebox.showwarning("Select Record", "Please select a record to delete")
            return

        tid = teacher_table.item(selected_item)["values"][0]  # Get Tid from the selected record

        conn = connect_db()
        cursor = conn.cursor()
        cursor.execute("DELETE FROM Teacher WHERE Tid = ?", (tid,))
        conn.commit()
        conn.close()
        messagebox.showinfo("Deleted", f"Teacher with ID {tid} deleted successfully")
        display_data()
    except Exception as e:
        messagebox.showerror("Error", f"Failed to delete teacher: {e}")


# Clear input fields
def clear_inputs():
    var_teacher_name.set("")
    var_teacher_designation.set("")
    var_teacher_salary.set("")
    var_teacher_deptid.set("")


# GUI Design
app = Tk()
app.title("Teacher Management System")
app.geometry("800x500")
app.configure(bg="#f0f0f0")

# Variables for entry fields
var_teacher_name = StringVar()
var_teacher_designation = StringVar()
var_teacher_salary = StringVar()
var_teacher_deptid = StringVar()

# Header Section
header_label = Label(app, text="Teacher Management System", bg="#3498db", fg="white", font=("Arial", 20,
"bold"))
header_label.pack(side=TOP, fill=X)

# Input Frame
input_frame = Frame(app, bg="lightgray", pady=10, padx=10)
input_frame.pack(side=LEFT, fill=Y, padx=20)

# Input Fields
Label(input_frame, text="Teacher Name:", bg="lightgray", font=("Arial", 12)).grid(row=0, column=0,
sticky=W, pady=5)
Entry(input_frame, textvariable=var_teacher_name, font=("Arial", 12)).grid(row=0, column=1, pady=5)

Label(input_frame, text="Designation:", bg="lightgray", font=("Arial", 12)).grid(row=1, column=0,
sticky=W, pady=5)
Entry(input_frame, textvariable=var_teacher_designation, font=("Arial", 12)).grid(row=1, column=1,
pady=5)

Label(input_frame, text="Salary:", bg="lightgray", font=("Arial", 12)).grid(row=2, column=0, sticky=W,
pady=5)
Entry(input_frame, textvariable=var_teacher_salary, font=("Arial", 12)).grid(row=2, column=1, pady=5)

Label(input_frame, text="Dept ID:", bg="lightgray", font=("Arial", 12)).grid(row=3, column=0, sticky=W,
pady=5)
Entry(input_frame, textvariable=var_teacher_deptid, font=("Arial", 12)).grid(row=3, column=1, pady=5)
```

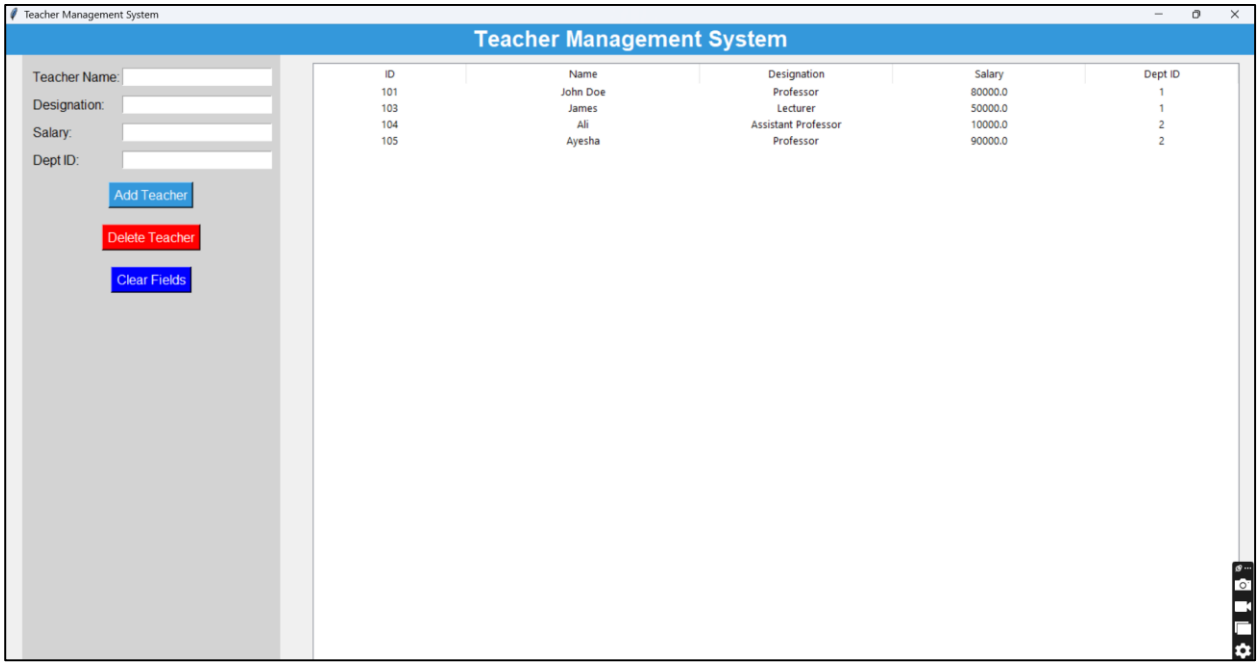
```
# Buttons
Button(input_frame, text="Add Teacher", command=add_teacher, bg="#3498db", fg="white", font=("Arial",
12)).grid(row=4, column=0, pady=10, columnspan=2)
Button(input_frame, text="Delete Teacher", command=delete_teacher, bg="red", fg="white", font=("Arial",
12)).grid(row=5, column=0, pady=10, columnspan=2)
Button(input_frame, text="Clear Fields", command=clear_inputs, bg="blue", fg="white", font=("Arial",
12)).grid(row=6, column=0, pady=10, columnspan=2)

# Table Frame
table_frame = Frame(app)
table_frame.pack(side=RIGHT, fill=BOTH, expand=True, padx=20, pady=10)

# Treeview Table
columns = ("ID", "Name", "Designation", "Salary", "Dept ID")
teacher_table = ttk.Treeview(table_frame, columns=columns, show="headings")
teacher_table.heading("ID", text="ID")
teacher_table.heading("Name", text="Name")
teacher_table.heading("Designation", text="Designation")
teacher_table.heading("Salary", text="Salary")
teacher_table.heading("Dept ID", text="Dept ID")

teacher_table.column("ID", width=50, anchor="center")
teacher_table.column("Name", width=150, anchor="center")
teacher_table.column("Designation", width=100, anchor="center")
teacher_table.column("Salary", width=100, anchor="center")
teacher_table.column("Dept ID", width=50, anchor="center")

teacher_table.pack(fill=BOTH, expand=True)

# Initialize Database and Display Data
create_tables()
display_data()

# Run the App
app.mainloop()
```

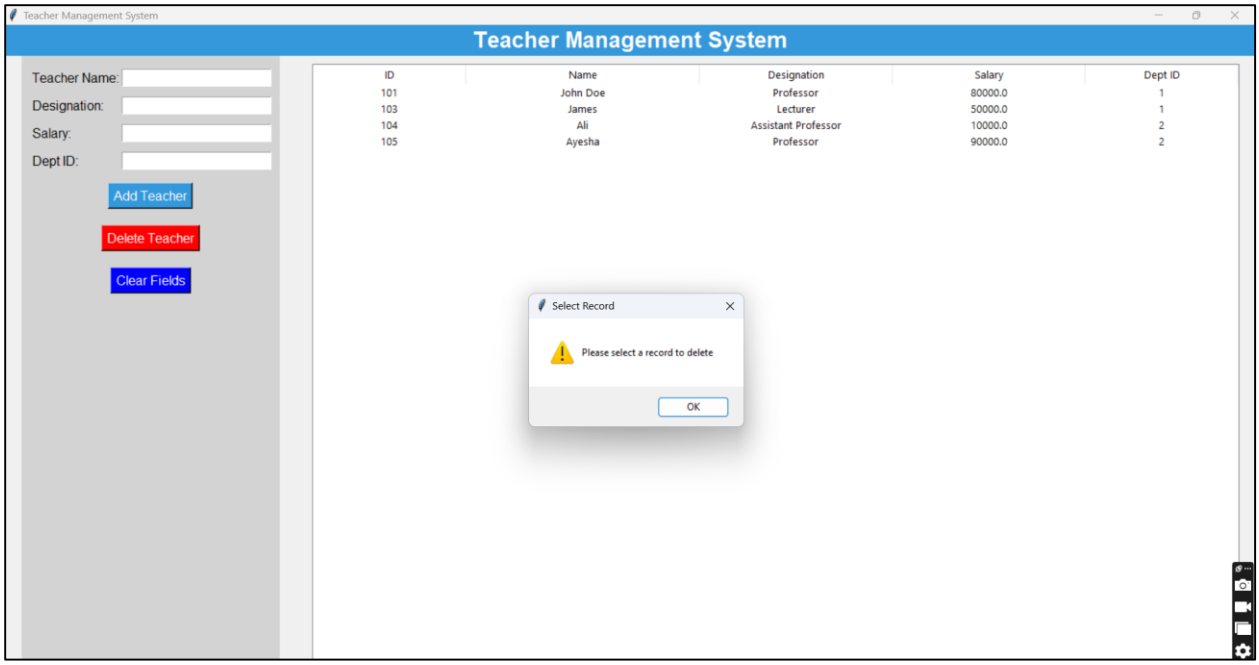
Output:

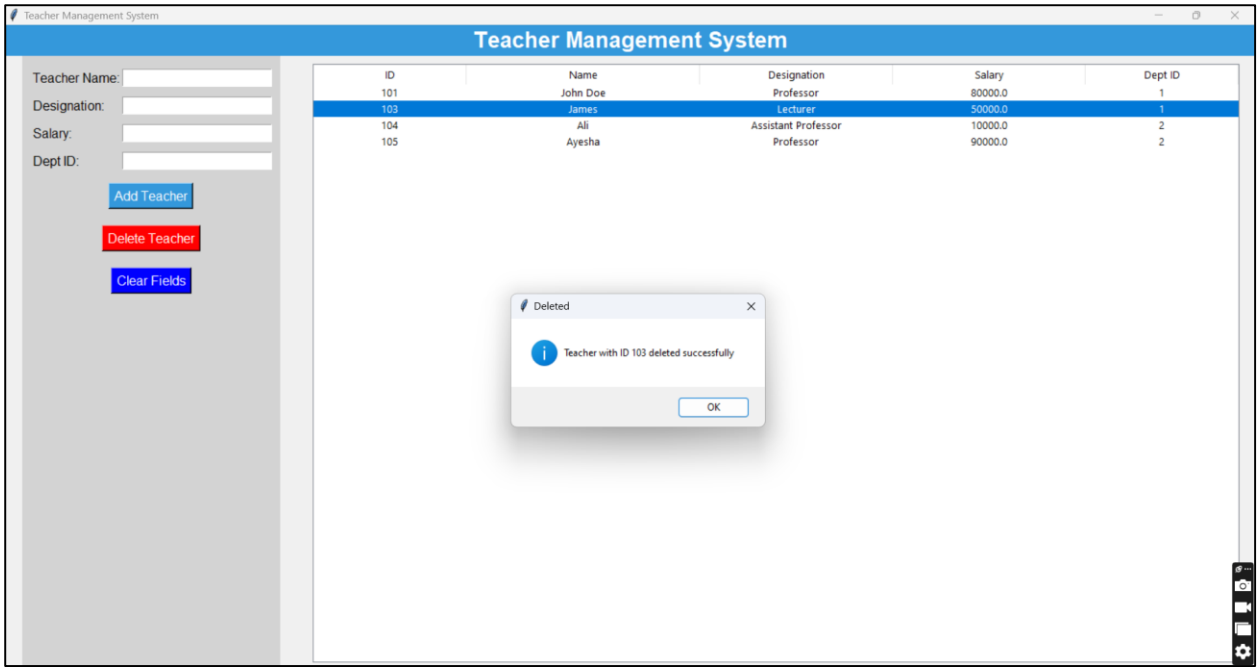

Teacher Management System
Teacher Management System
Teacher Name:
Designation:
Salary:
Dept ID:
Add Teacher
Delete Teacher
Clear Fields
ID
Name
Designation
Salary
Dept ID
101
John Doe
Professor
80000.0
1
103
James
Lecturer
50000.0
1
104
Ali
Assistant Professor
10000.0
2

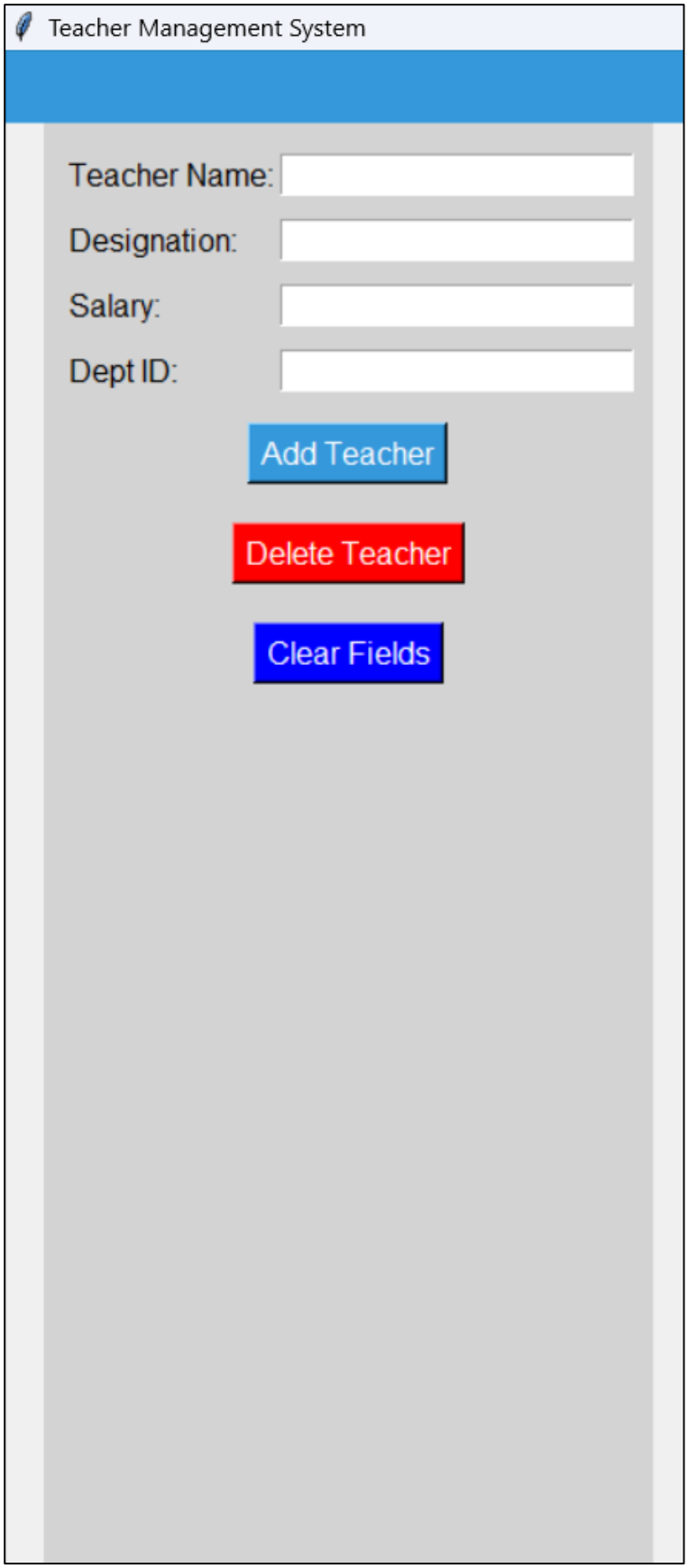

Teacher Management System
Teacher Management System
Teacher Name: Ayesha
Designation: Professor
Salary: 90000
Dept ID: 2
Add Teacher
Delete Teacher
Clear Fields
ID
Name
Designation
Salary
Dept ID
101
John Doe
Professor
80000.0
1
103
James
Lecturer
50000.0
1
104
Ali
Assistant Professor
10000.0
2

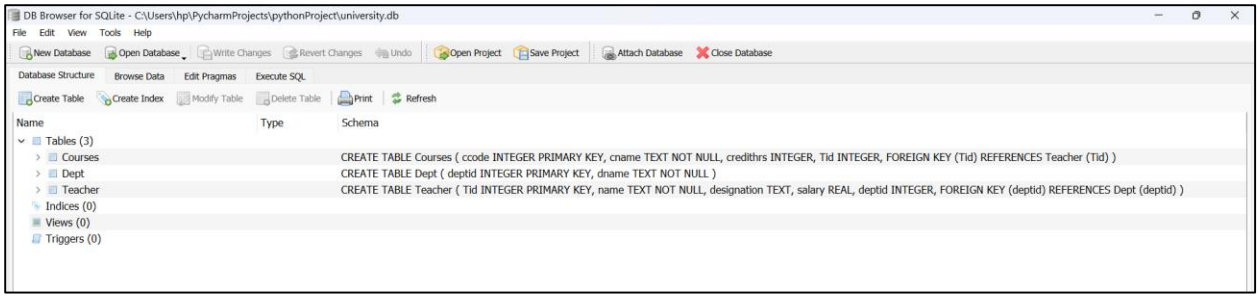

Teacher Management System
Teacher Management System
Teacher Name: Ayesha
Designation: Professor
Salary: 90000
Dept ID: 2
Add Teacher
Delete Teacher
Clear Fields
ID Name Designation Salary Dept ID
101 John Doe Professor 80000.0 1
103 James Lecturer 50000.0 1
104 Ali Assistant Professor 10000.0 2
Success
Teacher added successfully
OK

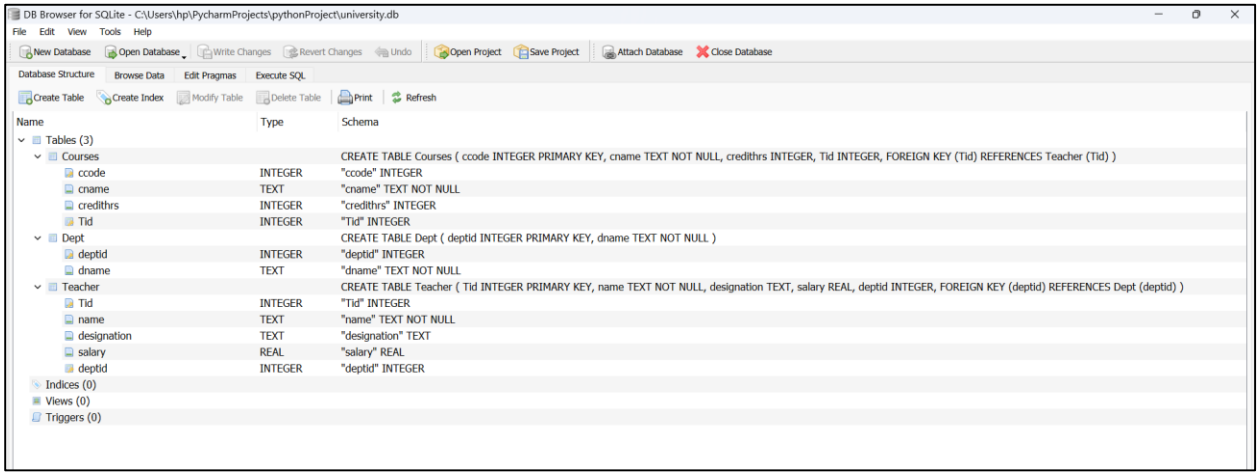

Teacher Management System
Teacher Management System
Teacher Name:
Designation:
Salary:
Dept ID:
Add Teacher
Delete Teacher
Clear Fields
ID Name Designation Salary Dept ID
101 John Doe Professor 80000.0 1
103 James Lecturer 50000.0 1
104 Ali Assistant Professor 10000.0 2
105 Ayesha Professor 90000.0 2


Teacher Management System
Teacher Management System
Teacher Name:
Designation:
Salary:
Dept ID:
Add Teacher
Delete Teacher
Clear Fields
ID Name Designation Salary Dept ID
101 John Doe Professor 80000.0 1
103 James Lecturer 50000.0 1
104 Ali Assistant Professor 10000.0 2
105 Ayesha Professor 90000.0 2
Select Record
Please select a record to delete
OK


Teacher Management System
Teacher Management System
Teacher Name:
Designation:
Salary:
Dept ID:
Add Teacher
Delete Teacher
Clear Fields
ID Name Designation Salary Dept ID
101 John Doe Professor 80000.0 1
103 James Lecturer 50000.0 1
104 Ali Assistant Professor 10000.0 2
105 Ayesha Professor 90000.0 2
Deleted
Teacher with ID 103 deleted successfully
OK


Teacher Management System
Teacher Name:
Designation:
Salary:
Dept ID:
Add Teacher
Delete Teacher
Clear Fields


DB Browser for SQLite - C:\Users\hp\PycharmProjects\pythonProject\university.db
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
Create Table Create Index Modify Table Delete Table Print Refresh
Name Type Schema
Tables (3)
Courses CREATE TABLE Courses (ccode INTEGER PRIMARY KEY, cname TEXT NOT NULL, credithrs INTEGER, Tid INTEGER, FOREIGN KEY (Tid) REFERENCES Teacher (Tid))
Dept CREATE TABLE Dept (deptid INTEGER PRIMARY KEY, dname TEXT NOT NULL)
Teacher CREATE TABLE Teacher (Tid INTEGER PRIMARY KEY, name TEXT NOT NULL, designation TEXT, salary REAL, deptid INTEGER, FOREIGN KEY (deptid) REFERENCES Dept (deptid))
Indices (0)
Views (0)
Triggers (0)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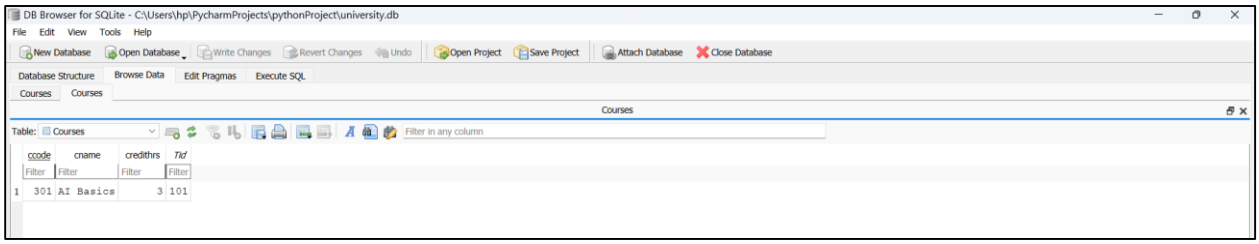
Tables (3)
Courses CREATE TABLE Courses (ccode INTEGER PRIMARY KEY, cname TEXT NOT NULL, credithrs INTEGER, Tid INTEGER, FOREIGN KEY (Tid) REFERENCES Teacher (Tid))
ccode INTEGER "ccode" INTEGER
cname TEXT "cname" TEXT NOT NULL
credithrs INTEGER "credithrs" INTEGER
Tid INTEGER "Tid" INTEGER
Dept CREATE TABLE Dept (deptid INTEGER PRIMARY KEY, dname TEXT NOT NULL)
deptid INTEGER "deptid" INTEGER
dname TEXT "dname" TEXT NOT NULL
Teacher CREATE TABLE Teacher (Tid INTEGER PRIMARY KEY, name TEXT NOT NULL, designation TEXT, salary REAL, deptid INTEGER, FOREIGN KEY (deptid) REFERENCES Dept (deptid))
Tid INTEGER "Tid" INTEGER
name TEXT "name" TEXT NOT NULL
designation TEXT "designation" TEXT
salary REAL "salary" REAL
deptid INTEGER "deptid" INTEGER
Indices (0)
Views (0)
Triggers (0)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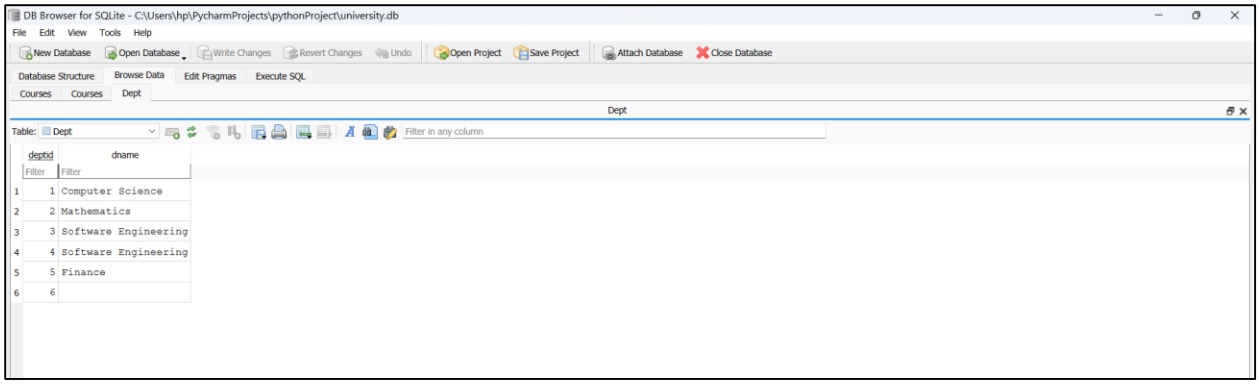
Database Structure Browse Data Edit Pragmas Execute SQL
Courses Courses
Courses
Table: Courses
Filter in any column
ccode cname credithrs Tid
Filter Filter Filter Filter
1 301 AI Basics 3 101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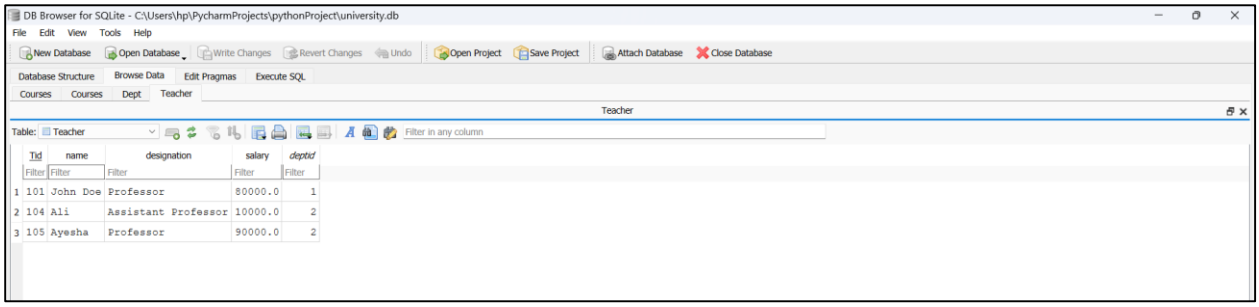
Database Structure Browse Data Edit Pragmas Execute SQL
Courses Courses Dept
Dept
Table: Dept
Filter in any column
deptid dname
Filter Filter
1 1 Computer Science
2 2 Mathematics
3 3 Software Engineering
4 4 Software Engineering
5 5 Finance
6 6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Courses Courses Dept Teacher
Teacher
Table: Teacher
Filter in any column
Tid name designation salary deptid
Filter Filter Filter Filter Filter
1 101 John Doe Professor 80000.0 1
2 104 Ali Assistant Professor 10000.0 2
3 105 Ayesha Professor 90000.0 2